POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 4031 | Semester IV | Theory | 4 Credits

Switch Gear and Protection

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electrical & Electronics Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 4031 as Switch Gear and Protection in Semester IV for Electrical & Electronics Engineering. The official SITTTR detailed source was unavailable during this cycle. With user approval, explanatory coverage uses reliable public diploma switchgear/protection curriculum sources and is not represented as recovered official SITTTR Course 4031 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders. This is a study handbook; switching, testing, relay setting and maintenance of live/high-energy systems require approved procedures, competent authorization and site-specific safety controls.

Verified source note: The official detailed source was inaccessible. Public sources supporting faults, fuses, breakers, relays, equipment protection and substation practice are linked in References.

COURSE DASHBOARD

What You Are Expected to Learn

60

Contact hours

4

Credits

Theory CIE 40

CIE

Theory ESE 60

ESE

Course introduction

Switch Gear and Protection develops the principles used to detect faults, isolate damaged sections and maintain continuity for healthy power-system equipment. It relates fault behaviour to fuses, circuit breakers, relays, instrument transformers, protection schemes, substations and safe maintenance practice.

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Prerequisite foundation

Electrical machines, basic power-system circuits, AC fundamentals and strict electrical-safety awareness.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain fault effects, protection requirements and basic switchgear functions.
2. Describe fuses, circuit breakers and arc-interruption principles.
3. Explain protective-relay types, characteristics and coordination concepts.
4. Explain protection of major equipment/feeders/busbars and safe substation practice.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

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Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain fault conditions, protection objectives and surge/overvoltage concepts.	Understand
CO2	Describe fuses and circuit breakers, their ratings, arc interruption and safe application.	Understand
CO3	Explain protective relays and selectivity/coordination principles for basic schemes.	Apply
CO4	Explain protection concepts for transformers, generators, feeders, busbars and substations.	Understand

Official CO-PO articulation matrix							
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	Title	Official major topics	Hours	CO	Handbook section
1	Protection Fundamentals and Faults	Need for protection, fault types/effects, short-circuit awareness, selectivity, reliability, sensitivity, speed, surge protection and earthing concepts.	Approx. 14 hours	CO1	Module 1
2	Fuses, Circuit Breakers and Switchgear	Switchgear functions, fuses/HRC fuses, breaker ratings, arcs, arc interruption and common LV/MV/HV breaker types.	Approx. 16 hours	CO2	Module 2
3	Protective Relays and Coordination	Relay terminology, electromagnetic/static/numerical relays, overcurrent, differential, distance and time-current coordination.	Approx. 16 hours	CO3	Module 3
4	Equipment Protection and Substation Practice	Generator/transformer/feeder/busbar protection, CT/PT, neutral earthing, substation layout, testing, maintenance and safety.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Protection Fundamentals and Faults

Need for protection, fault types/effects, short-circuit awareness, selectivity, reliability, sensitivity, speed, surge protection and earthing concepts.

CO1 · Approx. 14 hours

Fuses, Circuit Breakers and Switchgear

Switchgear functions, fuses/HRC fuses, breaker ratings, arcs, arc interruption and common LV/MV/HV breaker types.

CO2 · Approx. 16 hours

Protective Relays and Coordination

Relay terminology, electromagnetic/static/numerical relays, overcurrent, differential, distance and time-current coordination.

CO3 · Approx. 16 hours

Equipment Protection and Substation Practice

Generator/transformer/feeder/busbar protection, CT/PT, neutral earthing, substation layout, testing, maintenance and safety.

CO4 · Approx. 14 hours

MODULE 1

Protection Fundamentals and Faults

Need for protection, fault types/effects, short-circuit awareness, selectivity, reliability, sensitivity, speed, surge protection and earthing concepts.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

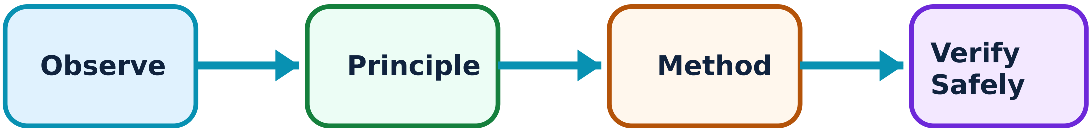
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Protection Fundamentals and Faults: identify the system, state its principle, follow the correct method and verify the result safely.

1. Protection purpose and qualities−

A protection system detects an abnormal condition and isolates only the affected section as quickly, selectively, reliably and safely as practical. It includes sensing, decision, tripping and interrupting elements.

Study and application method

For a simple one-line diagram, identify protected zone, likely faults, sensing quantity, trip device, primary protection and backup path.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: For a simple one-line diagram, identify protected zone, likely faults, sensing quantity, trip device, primary protection and backup path.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Protection design balances multiple requirements; fast operation alone is not sufficient if it unnecessarily disconnects healthy supply.

2. Faults and fault effects−

Short circuits and earth faults can create high currents, thermal/mechanical stress, voltage depression and instability. Fault studies estimate quantities needed to select and coordinate rated equipment.

Study and application method

Classify the fault, draw the affected current path and state which ratings/data a qualified study would require.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Classify the fault, draw the affected current path and state which ratings/data a qualified study would require.

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Common mistake / precaution: Do not calculate or apply fault levels without correct system base, impedance data, connection and current standard.

3. Surges, lightning and earthing−

Lightning and switching can cause overvoltages; arresters and grounding/earthing arrangements limit hazardous effects. Neutral earthing influences fault current and protection response.

Study and application method

Trace surge/fault-energy path to earth, identify protective device role and distinguish operational earthing from equipment bonding.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Trace surge/fault-energy path to earth, identify protective device role and distinguish operational earthing from equipment bonding.

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Common mistake / precaution: Earthing is safety-critical; never modify it or rely on a visual check instead of approved testing.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Fuses, Circuit Breakers and Switchgear

Switchgear functions, fuses/HRC fuses, breaker ratings, arcs, arc interruption and common LV/MV/HV breaker types.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

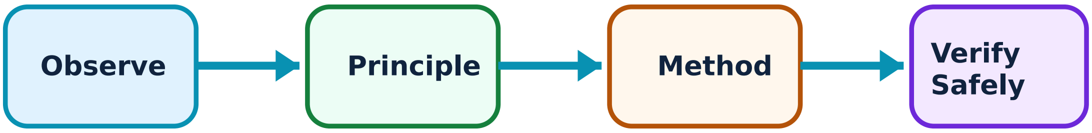
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Fuses, Circuit Breakers and Switchgear: identify the system, state its principle, follow the correct method and verify the result safely.

1. Fuses and switching devices−

A fuse interrupts current by melting a calibrated element; switchgear also includes isolators, switches, CT/PT, relays, arresters and breakers for control, isolation and protection.

Study and application method

Identify device function—control, isolation, sensing or interruption—and compare normal-current, fault-current and maintenance requirements.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify device function—control, isolation, sensing or interruption—and compare normal-current, fault-current and maintenance requirements.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Never replace a fuse with another rating or improvised conductor; protection coordination and fire safety can be defeated.

2. Circuit-breaker operation and ratings−

A circuit breaker opens a fault circuit using a trip mechanism and arc-extinction medium. Key ratings include voltage, continuous current, short-circuit breaking/making capacity and operating duty.

Study and application method

Read a device nameplate/data sheet, relate rating to system voltage/fault level/duty and state why coordinated protective tripping is needed.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Read a device nameplate/data sheet, relate rating to system voltage/fault level/duty and state why coordinated protective tripping is needed.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: A breaker must be selected and maintained for the actual system duty; a physical fit does not establish suitability.

3. Arc interruption and breaker types−

Oil, air, SF6 and vacuum technologies extinguish arcs by different mechanisms. Selection considers voltage level, interrupting duty, maintenance, environment and approved manufacturer guidance.

Study and application method

Compare medium, interruption principle, application and maintenance constraints in a table based on verified documentation.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare medium, interruption principle, application and maintenance constraints in a table based on verified documentation.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not open enclosures or perform breaker servicing without isolation, proving dead, lockout/tagout and authorisation.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Protective Relays and Coordination

Relay terminology, electromagnetic/static/numerical relays, overcurrent, differential, distance and time-current coordination.

Approx. 16 hours **CO3** **Understand · Apply**

Learning goals

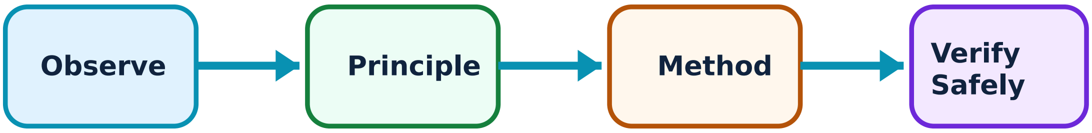
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Protective Relays and Coordination: identify the system, state its principle, follow the correct method and verify the result safely.

1. Relay functions and settings−

A relay measures a quantity such as current, voltage, impedance or differential current and issues a trip decision when its logic/setting criteria are met. Pickup, time setting and curve characteristics influence operation.

Study and application method

For a protection task, state measured quantity, protected zone, pickup/time rationale, breaker trip target and backup philosophy.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: For a protection task, state measured quantity, protected zone, pickup/time rationale, breaker trip target and backup philosophy.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Relay settings are controlled engineering data; never alter them without approved coordination study and authorization.

2. Overcurrent and differential protection−

Overcurrent protection operates when current/time thresholds are exceeded; differential protection compares currents entering/leaving a zone and is sensitive to internal faults when correctly designed.

Study and application method

Draw CT locations and relay zone, identify internal versus external fault current pattern and explain selectivity conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw CT locations and relay zone, identify internal versus external fault current pattern and explain selectivity conceptually.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: CT polarity, saturation and wiring errors can cause maloperation; testing must follow approved procedures.

3. Distance and numerical protection−

Distance protection estimates apparent impedance to identify a line fault zone. Numerical relays combine sensing, logic, communication and event recording, requiring configuration/version/control discipline.

Study and application method

Use a one-line diagram to identify zones and backup; document settings source, logic revision, test evidence and event-record interpretation path.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Use a one-line diagram to identify zones and backup; document settings source, logic revision, test evidence and event-record interpretation path.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Digital protection introduces cyber/configuration risk; access and change control are part of protection reliability.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Equipment Protection and Substation Practice

Generator/transformer/feeder/busbar protection, CT/PT, neutral earthing, substation layout, testing, maintenance and safety.

Approx. 14 hours | CO4 | Understand · Apply

Learning goals

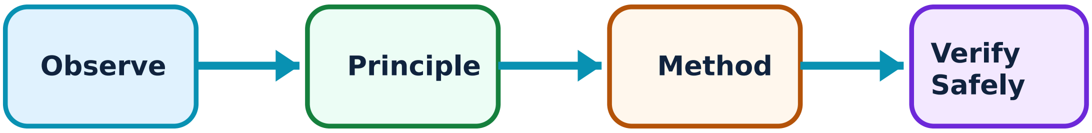
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Equipment Protection and Substation Practice: identify the system, state its principle, follow the correct method and verify the result safely.

1. Transformer and generator protection−

Transformers and generators face internal faults, winding/earth faults, overcurrent, overheating and abnormal operating conditions. Differential, Buchholz (where applicable), earth-fault and other schemes address defined risks.

Study and application method

Identify equipment, likely fault/abnormal condition, sensing device and trip/alarm response; separate internal from external fault protection.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify equipment, likely fault/abnormal condition, sensing device and trip/alarm response; separate internal from external fault protection.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□□□ □□□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□ □□□□□□□□□□□ □□□□□ □□□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Common mistake / precaution: Do not infer a complete protection scheme from one device; actual schemes depend on rating, connection and grid requirements.

2. Feeder and busbar protection−

Feeder schemes use time-graded overcurrent, distance or differential principles; busbar protection must act rapidly and selectively because bus faults can be severe.

Study and application method

Draw source-to-load zones and explain primary/backup grading at a high level with correct fault-direction awareness.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw source-to-load zones and explain primary/backup grading at a high level with correct fault-direction awareness.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Coordination must be verified by a qualified protection study, not generic time settings.

3. Substation safety, testing and maintenance−

Substations integrate switching, protection, measurement, grounding and control. Testing verifies equipment/relay performance; maintenance preserves reliability and safety through approved schedules and records.

Study and application method

Prepare a conceptual maintenance/test checklist covering permit, isolation, LOTO, prove-dead, earth, test equipment, result record and return-to-service checks.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Prepare a conceptual maintenance/test checklist covering permit, isolation, LOTO, prove-dead, earth, test equipment, result record and return-to-service checks.

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Common mistake / precaution: Never enter, test or maintain electrical equipment without competent authorization and applicable safety procedure.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Protection Fundamentals and Faults

Use the concept flow to revise observation, principle, method and verification.

Module 2: Fuses, Circuit Breakers and Switchgear

Use the concept flow to revise observation, principle, method and verification.

Module 3: Protective Relays and Coordination

Use the concept flow to revise observation, principle, method and verification.

Module 4: Equipment Protection and Substation Practice

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Draw a simple protected feeder one-line diagram showing relay, breaker, CT and protected zone.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Compare HRC fuse, MCB/MCCB and circuit-breaker functions from verified datasheets.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Interpret a conceptual time-current coordination sketch supplied by the instructor.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Prepare a safety-focused substation inspection checklist for a simulated case only.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure		
Component	**Official mark / conversion**	**Student evidence**
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Protection Fundamentals and Faults

Explain Protection purpose and qualities and state one correct method, application or precaution.−

A protection system detects an abnormal condition and isolates only the affected section as quickly, selectively, reliably and safely as practical. It includes sensing, decision, tripping and interrupting elements.

Answer framework: For a simple one-line diagram, identify protected zone, likely faults, sensing quantity, trip device, primary protection and backup path.

Important point: Protection design balances multiple requirements; fast operation alone is not sufficient if it unnecessarily disconnects healthy supply.

Explain Faults and fault effects and state one correct method, application or precaution.−

Short circuits and earth faults can create high currents, thermal/mechanical stress, voltage depression and instability. Fault studies estimate quantities needed to select and coordinate rated equipment.

Answer framework: Classify the fault, draw the affected current path and state which ratings/data a qualified study would require.

Important point: Do not calculate or apply fault levels without correct system base, impedance data, connection and current standard.

Explain Surges, lightning and earthing and state one correct method, application or precaution.−

Lightning and switching can cause overvoltages; arresters and grounding/earthing arrangements limit hazardous effects. Neutral earthing influences fault current and protection response.

Answer framework: Trace surge/fault-energy path to earth, identify protective device role and distinguish operational earthing from equipment bonding.

Important point: Earthing is safety-critical; never modify it or rely on a visual check instead of approved testing.

Module 2 — Fuses, Circuit Breakers and Switchgear

Explain Fuses and switching devices and state one correct method, application or precaution.−

A fuse interrupts current by melting a calibrated element; switchgear also includes isolators, switches, CT/PT, relays, arresters and breakers for control, isolation and protection.

Answer framework: Identify device function—control, isolation, sensing or interruption—and compare normal-current, fault-current and maintenance requirements.

Important point: Never replace a fuse with another rating or improvised conductor; protection coordination and fire safety can be defeated.

Explain Circuit-breaker operation and ratings and state one correct method,

application or precaution.−

A circuit breaker opens a fault circuit using a trip mechanism and arc-extinction medium. Key ratings include voltage, continuous current, short-circuit breaking/making capacity and operating duty.

Answer framework: Read a device nameplate/data sheet, relate rating to system voltage/fault level/duty and state why coordinated protective tripping is needed.

Important point: A breaker must be selected and maintained for the actual system duty; a physical fit does not establish suitability.

Explain Arc interruption and breaker types and state one correct method, application or precaution.−

Oil, air, SF6 and vacuum technologies extinguish arcs by different mechanisms. Selection considers voltage level, interrupting duty, maintenance, environment and approved manufacturer guidance.

Answer framework: Compare medium, interruption principle, application and maintenance constraints in a table based on verified documentation.

Important point: Do not open enclosures or perform breaker servicing without isolation, proving dead, lockout/tagout and authorisation.

Module 3 — Protective Relays and Coordination

Explain Relay functions and settings and state one correct method, application or precaution.−

A relay measures a quantity such as current, voltage, impedance or differential current and issues a trip decision when its logic/setting criteria are met. Pickup, time setting and curve characteristics influence operation.

Answer framework: For a protection task, state measured quantity, protected zone, pickup/time rationale, breaker trip target and backup philosophy.

Important point: Relay settings are controlled engineering data; never alter them without approved coordination study and authorization.

Explain Overcurrent and differential protection and state one correct method, application or precaution.−

Overcurrent protection operates when current/time thresholds are exceeded; differential protection compares currents entering/leaving a zone and is sensitive to internal faults when correctly designed.

Answer framework: Draw CT locations and relay zone, identify internal versus external fault current pattern and explain selectivity conceptually.

Important point: CT polarity, saturation and wiring errors can cause maloperation; testing must follow approved procedures.

Explain Distance and numerical protection and state one correct method, application or precaution.−

Distance protection estimates apparent impedance to identify a line fault zone. Numerical relays combine sensing, logic, communication and event recording, requiring configuration/version/control discipline.

Answer framework: Use a one-line diagram to identify zones and backup; document settings source, logic revision, test evidence and event-record interpretation path.

Important point: Digital protection introduces cyber/configuration risk; access and change control are part of protection reliability.

Module 4 — Equipment Protection and Substation Practice

Explain Transformer and generator protection and state one correct method, application or precaution.−

Transformers and generators face internal faults, winding/earth faults, overcurrent, overheating and abnormal operating conditions. Differential, Buchholz (where applicable), earth-fault and other schemes address defined risks.

Answer framework: Identify equipment, likely fault/abnormal condition, sensing device and trip/alarm response; separate internal from external fault protection.

Important point: Do not infer a complete protection scheme from one device; actual schemes depend on rating, connection and grid requirements.

Explain Feeder and busbar protection and state one correct method, application or precaution.−

Feeder schemes use time-graded overcurrent, distance or differential principles; busbar protection must act rapidly and selectively because bus faults can be severe.

Answer framework: Draw source-to-load zones and explain primary/backup grading at a high level with correct fault-direction awareness.

Important point: Coordination must be verified by a qualified protection study, not generic time settings.

Explain Substation safety, testing and maintenance and state one correct method, application or precaution.−

Substations integrate switching, protection, measurement, grounding and control. Testing verifies equipment/relay performance; maintenance preserves reliability and safety through approved schedules and records.

Answer framework: Prepare a conceptual maintenance/test checklist covering permit, isolation, LOTO, prove-dead, earth, test equipment, result record and return-to-service checks.

Important point: Never enter, test or maintain electrical equipment without competent authorization and applicable safety procedure.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Protection Fundamentals and Faults.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Fuses, Circuit Breakers and Switchgear.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Protective Relays and Coordination.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Equipment Protection and Substation Practice.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.

4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Need for protection, fault types/effects, short-circuit awareness, selectivity, reliability, sensitivity, speed, surge protection and earthing concepts..
2. Develop a complete answer or practical plan for: Switchgear functions, fuses/HRC fuses, breaker ratings, arcs, arc interruption and common LV/MV/HV breaker types..
3. Develop a complete answer or practical plan for: Relay terminology, electromagnetic/static/numerical relays, overcurrent, differential, distance and time-current coordination..
4. Develop a complete answer or practical plan for: Generator/transformer/feeder/busbar protection, CT/PT, neutral earthing, substation layout, testing, maintenance and safety..

LAST-MILE REVISION

Quick Revision

Module 1: Protection Fundamentals and Faults

Need for protection, fault types/effects, short-circuit awareness, selectivity, reliability, sensitivity, speed, surge protection and earthing concepts.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Fuses, Circuit Breakers and Switchgear

Switchgear functions, fuses/HRC fuses, breaker ratings, arcs, arc interruption and common LV/MV/HV breaker types.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Protective Relays and Coordination

Relay terminology, electromagnetic/static/numerical relays, overcurrent, differential, distance and time-current coordination.

- Match each topic to CO3.
- Redraw or rehearse one key method.

- Check one common mistake/precaution.

Module 4: Equipment Protection and Substation Practice

Generator/transformer/feeder/busbar protection, CT/PT, neutral earthing, substation layout, testing, maintenance and safety.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus

Term / topic	One-line meaning
Protection purpose and qualities	A protection system detects an abnormal condition and isolates only the affected section as quickly, selectively, reliably and safely as practical.
Faults and fault effects	Short circuits and earth faults can create high currents, thermal/mechanical stress, voltage depression and instability.
Surges, lightning and earthing	Lightning and switching can cause overvoltages; arresters and grounding/earthing arrangements limit hazardous effects.
Fuses and switching devices	A fuse interrupts current by melting a calibrated element; switchgear also includes isolators, switches, CT/PT, relays, arresters and breakers for control, isolation and protection.
Circuit-breaker operation and ratings	A circuit breaker opens a fault circuit using a trip mechanism and arc-extinction medium.
Arc interruption and breaker types	Oil, air, SF6 and vacuum technologies extinguish arcs by different mechanisms.
Relay functions and settings	A relay measures a quantity such as current, voltage, impedance or differential current and issues a trip decision when its logic/setting criteria are met.
Overcurrent and differential protection	Overcurrent protection operates when current/time thresholds are exceeded; differential protection compares currents entering/leaving a zone and is sensitive to internal faults when correctly designed.
Distance and numerical protection	Distance protection estimates apparent impedance to identify a line fault zone.
Transformer and generator protection	Transformers and generators face internal faults, winding/earth faults, overcurrent, overheating and abnormal operating conditions.
Feeder and busbar protection	Feeder schemes use time-graded overcurrent, distance or differential principles; busbar protection must act rapidly and selectively because bus faults can be severe.
Substation safety, testing and maintenance	Substations integrate switching, protection, measurement, grounding and control.

LEARNING RESOURCES

References

Recommended switchgear and protection references

1. V. K. Mehta and Rohit Mehta, Principles of Power System.
2. B. A. Oza et al., Power System Protection and Switchgear.
3. J. B. Gupta, Switchgear and Protection.

Approved public switchgear and protection sources

- http://www.sgpbellary.com/images/EEE/syllabus/sem5-eee/3-Switchgear_Protection.pdf
- https://blog.se.com/energy-management-energy-efficiency/electrical-safety/2021/03/01/all-you-need-to-know-about-switchgear-what-is-it-how-does-it-work-and-types/

These sources support the study handbook while official Course 4031 content is unavailable; they do not replace official syllabus requirements or authorised electrical-work procedures.